

Let us take a quick look at some of the recent developments and research happening in solar cells.

A Sharp panel for mobile devices

Following the success of its LROGC02 type panel, Sharp plans to launch more solar components for mobile applications. The five models they will be launching in the coming year will have 13 per cent efficiency and a low mounting depth of only 0.8 mm, which makes it convenient to mount onto any device. The solar panels are also double-wired to ensure that their performance is maintained even in the case of a cell breakage.

The panels will provide a range of outputs and voltages—single cell types with a 30mW power output at a voltage of 0.5V to models with 10 cells and a power output of 300mW at 5V—by varying the number and configuration of the individual polycrystalline solar cells.

A solar cell's efficiency is measured in terms of the amount of electricity generated per watt of incident solar light. An efficiency of 13 per cent is decent, considering that the highest known efficiency for practically usable solar cells is only around 20 per cent.

The mini solar cells can extend the operating life of devices like digital cameras, laptops, portable media players, pocket PCs, etc. For sensors and remote controls, the cells can even provide 100 per cent power.

New manufacturing process for higher efficiency

As mentioned earlier, the semiconductor is usually used as thin films or as wafers in a solar cell. A wafer is actually a single, large photovoltaic diode—and forms a significant part of a solar cell's cost.

Solaicx, a solar cell ingot maker, has developed a manufacturing process for wafers used in solar panels, which it claims results in higher efficiencies



SolarMagic by National Semiconductors

because it has been designed keeping in mind the special needs of the solar cell industry. This they claim is a better alternative than using production equipment originally designed for the semiconductor industry.

Basically, wafers are made by slicing ingots. Semiconductor ingots, when used for making solar cell wafers, have a resistivity that changes from one end of the ingot to the other. As a result, wafers cut from the ingot must be sorted based on their ability to convert sunlight into electricity. This variation in resistivity is the problem that Solaicx claims to have solved. Its ingots can be sawed into wafers that uniformly produce solar cells of the highest possible efficiency.

SolarMagic helps win over shade

Following its entry into the solar energy space, National Semiconductors launched an interesting technology called SolarMagic—a power optimiser that helps make the most of solar panels even under the shades of trees, power lines, chimneys, fallen leaves, etc.

Even a 10 per cent shade on an array can result in as much as 50 per cent energy loss. The company claims that its SolarMagic power optimisers can maximise the energy potential of each individual panel, so that as much as 57 per cent of the lost energy can be reclaimed.

SolarMagic won the Intersolar Award in Munich this year. While

the company has not released technical notes about this product, there is much speculation on the Internet forums about how they have achieved this optimisation.

One reasonable explanation reads: "This issue is well-known when many solar cells are wired in series to make

a module. If you wire 100 cells in series, each producing (say) 0.5 volt at 4 amps, you will have an array that produces 50 volts at 4 amps, i.e., 200 watts. But if just one cell is shaded (just the shadow of your hand), that individual cell produces high impedance, blocking the current. Then you get very little out of the entire panel. Modern panels generally incorporate bypass diodes, so if a small part of the panel is shaded, the current goes around the dark cells, and you still get reasonable power out of the panel. Other approaches at dealing with shading at increasingly larger scales are micro-inverters (like those made by Enphase), and independent Maximum Power Point Tracking (some inverters track their strings of modules separately)." Others wonder if there is some intelligence embedded into the panels.

A 'soft' side of the solar story

Solar cells and their efficiency are not only related to hardware. Software too plays a significant role in yield management and maximising the efficiency of panels.

Earlier this year, MKS Instruments Inc. launched the Genesis Solar Yield Management System to deliver yield solutions to solar cell manufacturers in a cost-effective and expandable platform. According to the press release, the patented solar domain specific data analysis and visualisation software is based on proven, reliable semiconductor and flat panel analytical tools. It allows users to quickly identify and